



CodeGalaxy - An Agentic AI Based Interactive Programming Learning Platform

ABSTRACT

Finding CodeGalaxy is an interactive and intelligent programming learning platform developed to improve the understanding of programming concepts among beginners using Agentic AI technology. Traditional programming education mainly focuses on textbooks, lectures, and coding exercises, which help students understand syntax and output but often fail to explain the internal execution of programs. Concepts such as loops, arrays, linked lists, recursion, algorithms, and memory allocation are difficult for beginners because they cannot visualize how programs work internally. To overcome these limitations, CodeGalaxy provides an AI-powered environment where users can write code in multiple programming languages such as Java, Python, and JavaScript, while Agentic AI analyzes, explains, and visualizes the complete execution process.

The main functionality of CodeGalaxy is powered by multiple AI agents that work together to provide an intelligent learning experience. The Code Analysis Agent examines user-written programs, detects syntax and logical errors, and provides improvement suggestions. The Execution Visualization Agent converts program execution into real-time visual representations by showing variable changes, loop iterations, condition checking, memory allocation, and data structure operations. The Learning Assistant Agent acts as a virtual programming mentor that answers questions, explains concepts, and provides personalized guidance. Additionally, the Assessment Agent evaluates coding skills, generates suitable challenges, identifies learning gaps, and recommends practice activities based on user performance.

CodeGalaxy also integrates gamification features such as coding challenges, levels, rewards, badges, experience points, and progress tracking to make programming learning more engaging. The Agentic AI system continuously monitors user progress and adapts the difficulty level according to individual learning ability. By combining artificial intelligence, real-time

visualization, and gamified learning, CodeGalaxy transforms traditional programming education into an interactive and personalized experience. The platform helps beginners develop strong conceptual understanding, improve problem-solving skills, and learn programming through intelligent AI-driven guidance rather than only memorizing code.

Roll No	Student's Name	Signature
2311cs040064	Harsh Kumar	
2311cs040003	Abhishek Pratap Kushwaha	
2311cs040006	Aditya Raj	
Reviewer Comments		

Guide Name & Sign

Reviewer Sign

AD-Coordinator Sign